

K Subset Sums I

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Consider the sums of all 2^n subsets of the given array (including the empty subset with sum equal to zero).

Your task is to find the k smallest subset sums.

Input

The first line has two integers n and k : the size of the array and the number of subset sums k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print k integers: the k smallest subset sums in increasing order.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k \leq \min(2^n, 2 \cdot 10^5)$
- $-10^9 \leq x_i \leq 10^9$

Example

Input	Output
4 9 1 6 3 -3	-3 -2 0 0 1 1 3 3 4